

- 6 Fig. 6.1 shows a circuit consisting of two resistors connected in series to a d.c. supply.

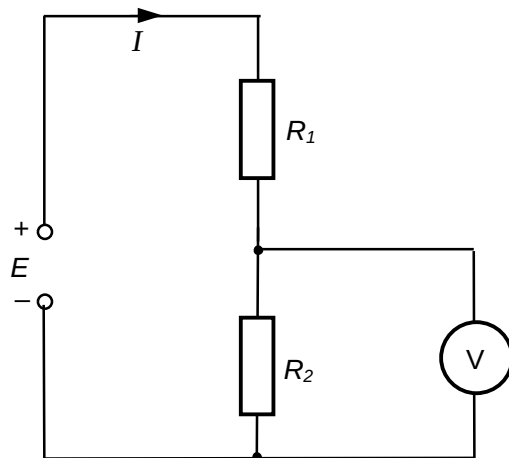


Fig. 6.1

The resistors have resistances R_1 and R_2 . The supply has e.m.f. E and negligible internal resistance. The current from the supply is I . The voltmeter has an infinite resistance.

- (a) By reference to Ohm's Law, show that the voltmeter reading V is given by the relation

$$V = \left(\frac{R_2}{R_1 + R_2} \right) E.$$

By Ohm's Law, $E = IR_{tot}$

$$I = \frac{E}{R_{tot}}$$

$$= \frac{E}{R_1 + R_2}$$

Potential difference across R_2 , $V = IR_2$

$$= \left(\frac{E}{R_1 + R_2} \right) R_2$$

$$= \left(\frac{R_2}{R_1 + R_2} \right) E \quad [\text{shown}]$$

(b) Fig. 6.2 shows a circuit that includes a thermistor.

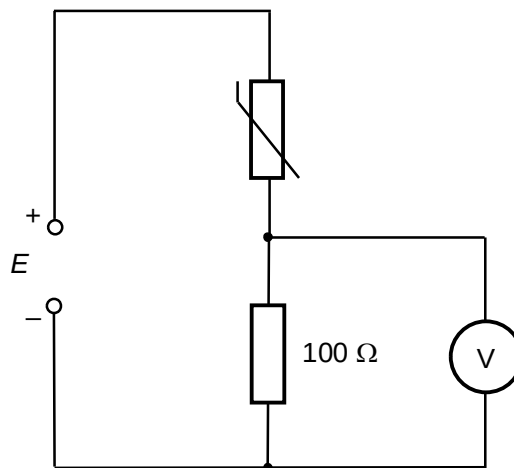


Fig. 6.2

Describe and explain how the voltmeter reading changes as the temperature of the thermistor is increased. [2]

As the temperature of the thermistor is increased, the resistance of the thermistor will decrease (R_1 decreases). As the resistance of the $100\ \Omega$ resistor remains constant, the ratio of its resistance to the total resistance will increase, i.e. $R_2/(R_1$

+ R_2) will increase. As such the potential difference across the $100\ \Omega$ resistor increases and the voltmeter will register a higher reading.

- (c) Fig. 6.3 shows a circuit that includes a potentiometer wire. It is used to test E_T , the e.m.f. of the test cell and the balance length is found to be b .

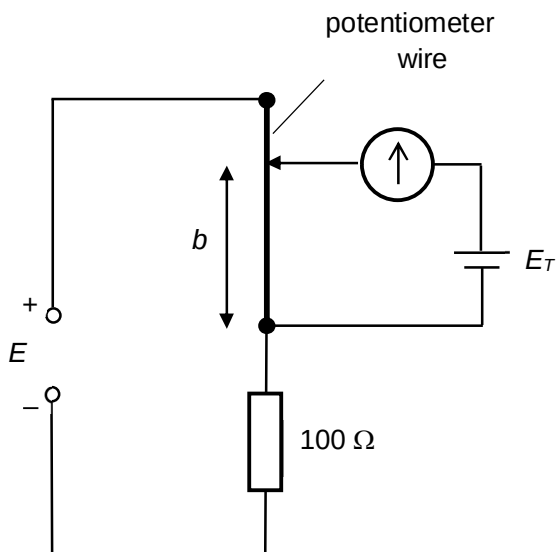


Fig. 6.3

- (i) The potentiometer wire, of 1.00 m long, is made of nichrome wire of resistivity $1.57 \times 10^{-6}\ \Omega\text{ m}$ and has a uniform radius of 0.10 mm. Determine the resistance of the potentiometer wire. [2]

$$\begin{aligned} R &= \frac{\rho l}{A} \\ &= \frac{\rho l}{\pi r^2} \\ &= \frac{1.57 \times 10^{-6} (1.00)}{\pi (0.10 \times 10^{-3})^2} \\ &= 50\ \Omega \end{aligned}$$

- (ii) Given that the ratio of $\frac{E_T}{E} = 0.20$, determine the balance length b . [2]

$$\text{Potential across potentiometer wire} = \frac{50}{50 + 100} \times E = 0.333\ E$$

$$\text{balance length } b = \frac{0.20}{0.333} \times 1.00 = 0.60\text{ m}$$